

Micro2: Expenditure Minimization, Duality, and Comparative Statics

Session 1: Expenditure Minimization, Hicksian Demand, and Concavity

1.1 Expenditure Minimization Problem (EMP)

Given prices $p_1, \dots, p_n > 0$ and a utility target $\bar{u} > u(0, \dots, 0)$, the consumer chooses $x \in \mathbb{R}_+^n$ to minimize expenditure subject to reaching the target:

$$\min_{x_1, \dots, x_n} \sum_{i=1}^n p_i x_i \quad (1)$$

$$\text{s.t. } u(x_1, \dots, x_n) \geq \bar{u}, \quad x_i \geq 0 \quad (i = 1, \dots, n). \quad (2)$$

A standard approach is to rewrite as a maximization of the negated objective and apply KKT.

1.2 Solving EMP via KKT

Write EMP in standard KKT form:

$$\max_{x \geq 0} - \sum_{i=1}^n p_i x_i \quad \text{s.t.} \quad u(x) - \bar{u} \geq 0.$$

Form the Lagrangian with one multiplier for the utility constraint and multipliers for nonnegativity constraints, then solve by cases (binding vs slack constraints).

1.3 Worked example: quasi-linear target constraint

For $p_1, p_2 > 0$ and target $\bar{u}$, consider

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \quad \text{s.t.} \quad x_1 + \ln x_2 \geq \bar{u}, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

Using KKT and case analysis yields a piecewise solution:

$$(x_1^*, x_2^*) = \begin{cases} \left(\bar{u} - \ln\left(\frac{p_1}{p_2}\right), \frac{p_1}{p_2} \right), & \text{if } \bar{u} \geq \ln\left(\frac{p_1}{p_2}\right), \\ (0, e^{\bar{u}}), & \text{if } \bar{u} < \ln\left(\frac{p_1}{p_2}\right). \end{cases}$$

(Feasibility requires $x_2 > 0$ because $\ln x_2 \rightarrow -\infty$ as $x_2 \rightarrow 0$.)

2.1 Hicksian demand and expenditure function

- **Hicksian demand correspondence** $h(p, \bar{u})$: the set of EMP minimizers at prices p and target $\bar{u}$.
- **Expenditure function** $e(p, \bar{u})$: minimized expenditure at $(p, \bar{u})$, i.e.

$$e(p, \bar{u}) = \min\{p \cdot x : u(x) \geq \bar{u}, x \geq 0\}.$$

2.2 Properties (under continuity and monotonicity of u)

Key properties emphasized:

- **Hicksian homogeneity (degree 0 in prices)**: $h(\alpha p, \bar{u}) = h(p, \bar{u})$ for $\alpha > 0$.
- **Target is met at optimum**: for any $x^* \in h(p, \bar{u})$, $u(x^*) = \bar{u}$.
- **Expenditure homogeneity (degree 1 in prices)**: $e(\alpha p, \bar{u}) = \alpha e(p, \bar{u})$.
- **Monotonicity**: e increases in $\bar{u}$ and is non-decreasing in each p_i .
- **Concavity in prices**: $e(\cdot, \bar{u})$ is concave in p .

For the quasi-linear example, both $h(p, \bar{u})$ and $e(p, \bar{u})$ are computed in closed form (piecewise).

3–4 Concavity tools and why KKT points solve the problems

- Definitions of convex sets and concave functions; concavity can be checked via the Hessian when differentiable.
- A twice-differentiable function is concave iff its Hessian is **negative semi-definite** ($z^\top H z \leq 0$ for all z).
- **Concave programming principle**: if the objective and constraints are concave (in the appropriate sense), then any KKT point is a global solution.
- Practical criteria for negative (semi-)definiteness: eigenvalues; determinant-based tests (with cautions beyond 2×2).

Session 2: Duality, Envelope Theorem, Shephard/Roy, and Slutsky

1 Duality between UMP and EMP

Assume $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ is continuously differentiable and increasing.

- If x^* solves the **utility maximization problem** (UMP) at (p, m) , then it also solves EMP at prices p with target $\bar{u} = v(p, m)$.
- If x^* solves EMP at $(p, \bar{u})$, then it also solves UMP at prices p with budget $m = e(p, \bar{u})$.

Core identities linking Marshallian demand $x(\cdot)$, Hicksian demand $h(\cdot)$, indirect utility $v(\cdot)$, and expenditure $e(\cdot)$:

- $x(p, m) = h(p, v(p, m))$
- $h(p, \bar{u}) = x(p, e(p, \bar{u}))$
- $e(p, v(p, m)) = m$
- $v(p, e(p, \bar{u})) = \bar{u}$

2 Envelope theorem (parameter derivatives of the value function)

For a parametric constrained maximization problem, the derivative of the optimal value with respect to a parameter equals the partial derivative of the Lagrangian with respect to that parameter, evaluated at the optimum (terms involving $\frac{\partial x^*}{\partial q}$ drop out by FOCs and complementary slackness).

3 Applications of the envelope theorem

- Shephard's lemma (EMP):

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

- Roy's identity (UMP):

$$x_i(p, m) = -\frac{\partial v(p, m)/\partial p_i}{\partial v(p, m)/\partial m}.$$

4 Slutsky equation and decomposition of price effects

Using the duality relation $h(p, \bar{u}) = x(p, e(p, \bar{u}))$ and applying derivatives:

$$\frac{\partial h_i(p, \bar{u})}{\partial p_j} = \frac{\partial x_i(p, m)}{\partial p_j} + \frac{\partial x_i(p, m)}{\partial m} x_j(p, m), \quad \text{with } m = e(p, \bar{u}).$$

Rearranging gives the standard decomposition:

$$\frac{\partial x_i(p, m)}{\partial p_j} = \underbrace{\frac{\partial h_i(p, \bar{u})}{\partial p_j}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_i(p, m)}{\partial m} x_j(p, m)}_{\text{income effect}}.$$

4.2 Slutsky matrix

Define the Slutsky matrix $S(p, m)$ with entries

$$S_{ij}(p, m) = \frac{\partial x_i(p, m)}{\partial p_j} + \frac{\partial x_i(p, m)}{\partial m} x_j(p, m).$$

Equivalently, $S_{ij}(p, m) = \frac{\partial h_i(p, v(p, m))}{\partial p_j}$, and by Shephard's lemma,

$$S(p, m) = \left[\frac{\partial^2 e(p, v(p, m))}{\partial p_i \partial p_j} \right]_{i,j}.$$

Since $e(\cdot, \bar{u})$ is concave in prices, $S(p, m)$ is **symmetric** and **negative semi-definite**.